→ Cybernetic models: Pyruvate - Fumerate

→ Hybrid cybernetic model → metabolic networks
(PSSH for intracellular metabolites)

www.expasy.org (Structure of metabolic networks)

→ Elementary mode

→ Linear programming → deal w/ max. a linear functional
maximizing
↳ no of feasible soln are on a convex polygon
(solutions are hopping from one vertex to another)

Choosing which path (by metabolizing) → maximizing the final output (Total carbon intake, or max. viable etc.)

Convex comb.
↳ Linear comb. w/ positive coefficients

↓
Flux space and yield space

→ FBA solution vs. cybernetic model
(Paulsen)

→ $n(m,t)$ ← No. density — Total number density

$N(t) = \int_0^\infty n(m,t)\,dm$ } All cells in the interval

→ $r(m,s)$ ← Rate of increase of cell mass



Flux $= r(a,s) \cdot n(a,t)$

→ Division rate of a cell:
($f(m,s)$) → Fraction of cells of mass m dividing

→ Daughter cell Mass dist: [contribute differently to the mass]

$p(m|m',s)\,dm$ → fraction of daughter cells formed from mother cells of mass m' (w/ mass b/w m & $m+dm$)
↑ given a known certain event
(independent vs. dependent event)

Population balance Equation:

no. of cells of mass m b/w a & b

$\frac{d}{dt}\int_a^b n(m,t)\,dm$ = Rate of increase of cells of mass m

m and t are independent vars (can take $\frac{d}{dt}$ inside)

= Rate of cells entering at a − Rate leaving @ b − rate of cells lost from (a,b) from division + Rate of cells arriving by division of larger cells

$r(a,s)\cdot n(a,t)$ — $r(b,s)\,n(b,t)$ — $\int_a^b f(m,s)\,n(m,t)\,dm$ — $\int_a^b dm \int_m^{2} \frac{1}{m'} f(m',s)\,n(m',t)\,dm'\,p(m|m',s)$

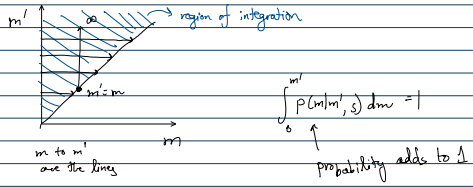
2 cells

r @ $m=0$ is 0

at any given time $n \neq \infty$; i.e. @ $m=\infty$, $n=0$

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$$\int_0^{m'} p(m|m',s)\,dm = 1$$

↑ probability adds to 1

$$\therefore \frac{dN}{dt} = \int_0^{\infty} \Gamma(m,s)\, n(m,t)\, dm$$

$$c(t) = \int_0^{\infty} m \cdot n(m,t)\, dm$$

↑ Concentration (biomass)

+ Biomass balance for lumped biophase:

$$+ \frac{ds}{dt} = -\frac{1}{r}\int_0^{\infty} dm\; n(m,t)\, r(m,s)$$

Dissolution process

Y → conc. of solute (in bulk)

Y^* → solubility

$$\frac{dN}{dt} = -k' \lim_{m\to\infty}\left[m^{2/3}\, n(m,t)\right]$$

↑ negative because losing mass

Internal coordinate change → due to reaction

$S(x)$ → differential interface surface.

Method of characteristics [hyperbolic vs. elliptical?]

2 cells from each method